

AI-Powered Trade Compliance and Risk Decision Fabric

Executive Summary

Architected and delivered a cloud-native, distributed AI-powered trade surveillance and regulatory compliance platform engineered for high-frequency capital market environments. Designed an event-driven microservices ecosystem leveraging Apache Kafka, Kubernetes, distributed streaming inference pipelines, and graph-based entity correlation to detect anomalous trading behaviors, market manipulation patterns, insider trading signals, and regulatory violations in near real time.

Business Impact

Metric	Realistic Target	Rationale
Manual screening effort reduction	35% to 50%	Achievable when low-risk auto-clear rules are stable and document extraction quality is above baseline
False positive reduction in sanctions/entity screening	20% to 35%	Dependent on entity resolution quality and alias graph enrichment

Analyst turnaround time improvement	30% to 45%	Case prep is accelerated by pre-assembled evidence and generated rationale
Straight-through processing rate	40% to 60% for low-risk flows	Only for bounded transaction classes with mature rule coverage
Audit preparation time	50% to 70% reduction	Centralized evidence capture and immutable decision traces materially reduce reconstruction effort
Model-assisted case summarization time	60% to 75% reduction	RAG-generated investigation packs reduce manual note collation

Problem Statement

Global trade operations are constrained by fragmented regulatory data, volatile sanction regimes, heterogeneous shipping documents, counterparty opacity, and delayed exception handling. Manual review chains are too slow for modern transaction velocity and too brittle for rapidly changing geopolitical risk. The net

effect is operational latency, elevated false positives in screening, inconsistent analyst decisions, and material exposure to compliance breaches.

For a platform such as TradeGuardAI, the core architectural problem is not merely building a document AI or a rules engine. The real challenge is constructing a distributed decisioning system that can ingest multimodal trade evidence, fuse deterministic compliance policies with probabilistic ML outputs, maintain full decision traceability, and execute near-real-time risk scoring without violating enterprise security, auditability, or jurisdictional controls.

Business Objective

The business objective is to operationalize a machine-assisted trade governance platform that reduces manual review load while increasing the precision, explainability, and turnaround time of trade transaction screening. The platform should support pre-trade, in-flight, and post-trade controls across customers, counterparties, commodities, routes, vessels, ports, invoices, letters of credit, bills of lading, and supporting trade documentation.

A production-grade target state should deliver:

- centralized ingestion of structured and unstructured trade artifacts,
- near-real-time compliance screening and risk scoring,
- human-in-the-loop adjudication for high-risk or low-confidence cases,
- immutable audit trails for regulatory defensibility,

- enterprise deployment primitives for scale, resiliency, and policy isolation.

Why Azure Is the Most Suitable Cloud

Azure is the most suitable primary infrastructure for this workload when the target customer profile includes regulated enterprises, BFSI trade desks, and organizations that need strong identity integration with Microsoft ecosystems, granular data governance, and managed AI services under a controlled enterprise posture. Azure AI Document Intelligence, Azure OpenAI, Microsoft Entra ID, Purview, Event Hubs, AKS, and Defender for Cloud combine well for a compliance-first platform where auditability and policy enforcement are as important as model capability.

The recommendation is therefore **Azure-first with optional multi-cloud extension points**. Multi-cloud portability can be retained at the container, data-contract, and model-serving layers, but the control plane should remain opinionated to avoid unnecessary operational entropy.

Tech Stack

Layer	Used Stack
Front-end	TypeScript
API & Edge	Azure Front Door, WAF, Azure API Management
Microservices	Python FastAPI, .NET for policy-heavy services, gRPC for internal low-latency calls

Orchestration	Temporal or Durable Functions for workflow execution
Eventing	Azure Event Hubs, Service Bus
Document Processing	Azure AI Document Intelligence, custom parsers, OCR fallback pipelines
ML/AI	Azure Machine Learning, XGBoost/LightGBM for tabular risk, PyTorch for graph or anomaly models, Azure OpenAI for controlled reasoning
Retrieval	Azure AI Search or dedicated vector store, graph database, hybrid retrieval orchestration
Data Stores	Azure SQL, Cosmos DB, Blob Storage, Redis
Graph Layer	Neo4j on AKS or equivalent managed graph-capable platform
Observability	OpenTelemetry, Prometheus, Grafana, Azure Monitor, Log Analytics
Security	Entra ID, Key Vault, Defender for Cloud, Purview, Sentinel

DevSecOps	GitHub Actions or Azure DevOps, Terraform, Helm, Trivy, Checkov, policy-as-code
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Resources & Hardware Requirements

Component	Baseline Sizing	Notes
AKS system pool	3 x 4 vCPU / 16 GB nodes	Control add-ons, ingress, observability
AKS app pool	4 to 6 x 8 vCPU / 32 GB nodes	Core microservices, orchestration, API workloads
GPU pool	Optional 1 to 2 x NVIDIA L4/T4 class nodes	Only if local embedding/inference is required instead of fully managed AI endpoints
Azure SQL	Business Critical 4 to 8 vCores	Transactional integrity, case management, rule metadata
Cosmos DB	10k to 30k RU/s equivalent starting band	Screening events, flexible entities, variable throughput

Blob Storage	Hot + Cool tier lifecycle policy	Raw docs, parsed artifacts, retention evidence
Redis	2 to 3 node managed cache, 6 to 13 GB	Session, rule cache, feature cache
Event Hubs	2 to 4 throughput units to start	Burst-driven ingestion
Log Analytics	Sized for high-cardinality telemetry	Retention tuned by compliance class

Cost Optimization Techniques

- Separate synchronous decision paths from asynchronous enrichment paths so expensive AI and graph operations are executed only when confidence bands or policy triggers justify them.
- Use managed OCR and LLM endpoints selectively, not as the default for every transaction; deterministic parsers and cached templates should handle recurrent document layouts first.
- Apply autoscaling on AKS user pools using queue depth, event lag, and CPU/memory saturation rather than static overprovisioning.
- Route embeddings, batch enrichment, backfills, and graph recomputation to spot or scheduled compute windows where business SLAs permit.

- Implement document lifecycle policies across hot, cool, and archive tiers with retention classes mapped to regulatory requirements.
- Cache sanctions lists, entity profiles, route intelligence, and normalized trade taxonomies aggressively to reduce repeated external lookups.
- Use confidence-gated human review queues to minimize analyst workload inflation caused by low-quality model thresholds.

Challenges and Resolution Strategies

1. Ambiguous entity matching

Names of counterparties, vessels, and beneficial owners often appear in inconsistent forms across trade artifacts. This is addressed through probabilistic entity resolution, alias dictionaries, transliteration normalization, and graph-based evidence scoring rather than exact-name matching.

2. Unstructured document variability

Bills of lading, invoices, and letters of credit vary substantially across issuers and geographies. The mitigation pattern combines OCR, layout-aware extraction, template libraries, fallback parsers, and analyst feedback loops for schema drift correction.

3. Regulatory explainability

Pure black-box scoring is not acceptable in compliance workflows. Every decision therefore needs rule trace, feature attribution, source evidence references,

prompt/response logging for LLM usage, and policy-bounded natural language rationale generation.

4. Latency under mixed workloads

Document-heavy cases can stall lightweight API decisions if both share compute pools. The resolution is workload isolation: synchronous APIs on dedicated pools, asynchronous enrichment on queue-backed workers, and circuit breakers around optional enrichment stages.

5. Hallucination risk in AI-generated rationale

LLM output cannot be treated as authoritative. The mitigation is retrieval-bounded prompting, evidence citation, policy guardrails, output schema enforcement, and a hard rule that final compliance disposition is produced by deterministic orchestration, not free-form generative output.

Security & Compliance Implementations

- Entra ID backed authentication with conditional access, MFA, and role-segregated RBAC for analysts, supervisors, auditors, and administrators.
- Private networking for data services, restricted egress, and mTLS between east-west service paths.
- Customer-managed keys in Key Vault for encryption-sensitive data domains.
- Tokenization or field-level encryption for PII and commercially sensitive transaction attributes.

- Full audit chain including input payload hash, document checksum, model version, prompt template version, retrieval context identifiers, rule set version, and final adjudication actor.
- Policy-as-code enforcement in CI/CD for network posture, secret exposure, image vulnerabilities, and IaC drift.
- Data classification, lineage, retention, and legal hold managed through Purview-aligned governance patterns.
- Continuous threat detection via Defender and SIEM correlation in Sentinel.

Roles and Responsibilities — Principal AI Architect (AI Platforms & Distributed Systems)

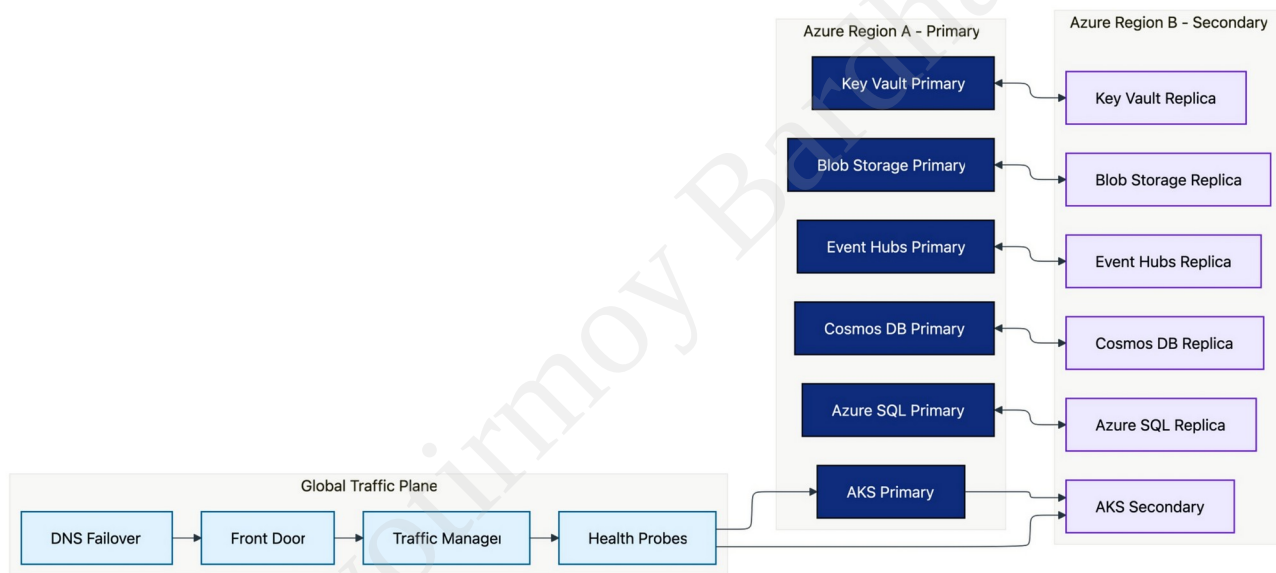
- Own the target-state platform architecture spanning ingestion, decisioning, AI/ML, graph intelligence, observability, and enterprise integration boundaries.
- Define reference patterns for deterministic rules, probabilistic models, and LLM-assisted reasoning so that each is used in the correct control envelope.
- Establish canonical domain models for transaction, counterparty, vessel, commodity, document, and screening decision entities.
- Drive cloud topology decisions, tenancy boundaries, workload isolation strategy, HA/DR posture, and scalability envelopes.

- Set engineering standards for API contracts, event schemas, idempotency, replay handling, trace propagation, and failure semantics.
- Govern model lifecycle architecture including feature generation, evaluation benchmarks, drift monitoring, champion-challenger rollout, and rollback controls.
- Design human-in-the-loop adjudication patterns that preserve analyst authority while maximizing automation value.
- Lead threat modeling, security architecture reviews, and compliance-by-design controls for sensitive trade intelligence workflows.
- Conduct critical design reviews for performance bottlenecks, data consistency risks, and cross-service coupling.
- Translate executive business goals into delivery-ready architecture increments and measurable technical success criteria.

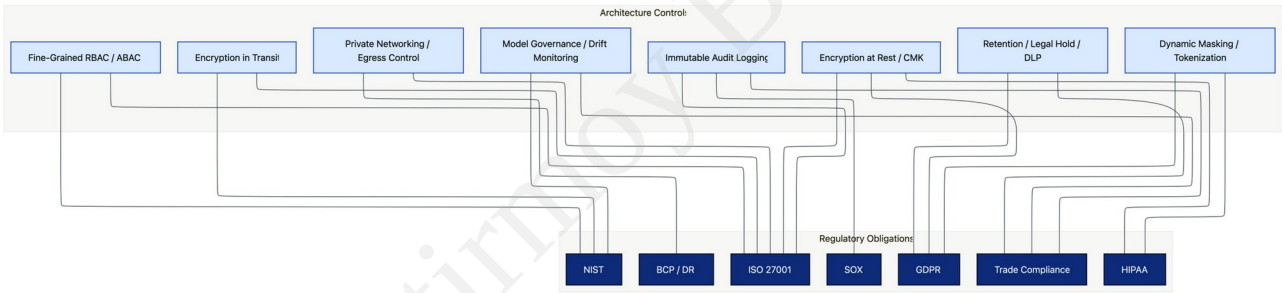
Appendix: Diagrams

DR-Topology

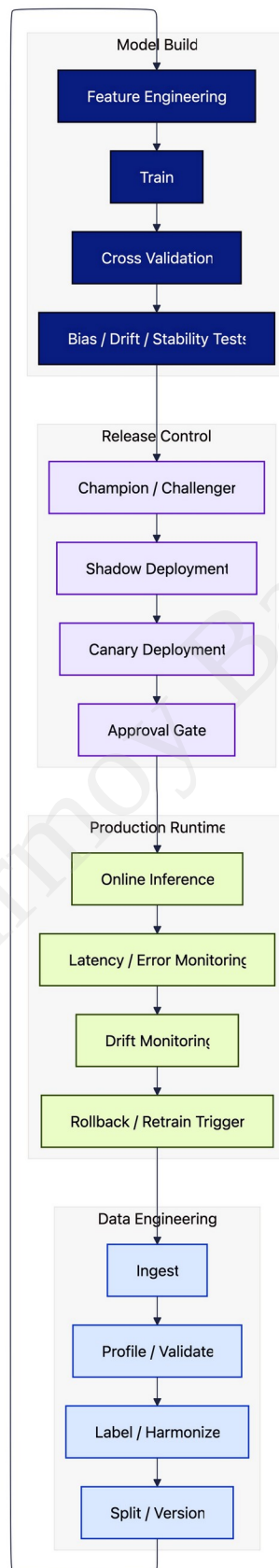
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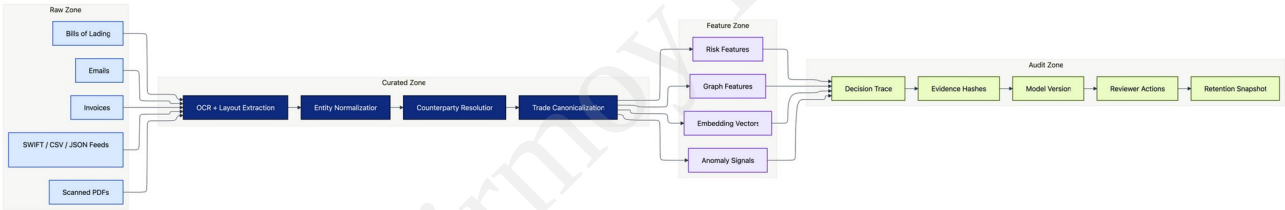
Control-Matrix



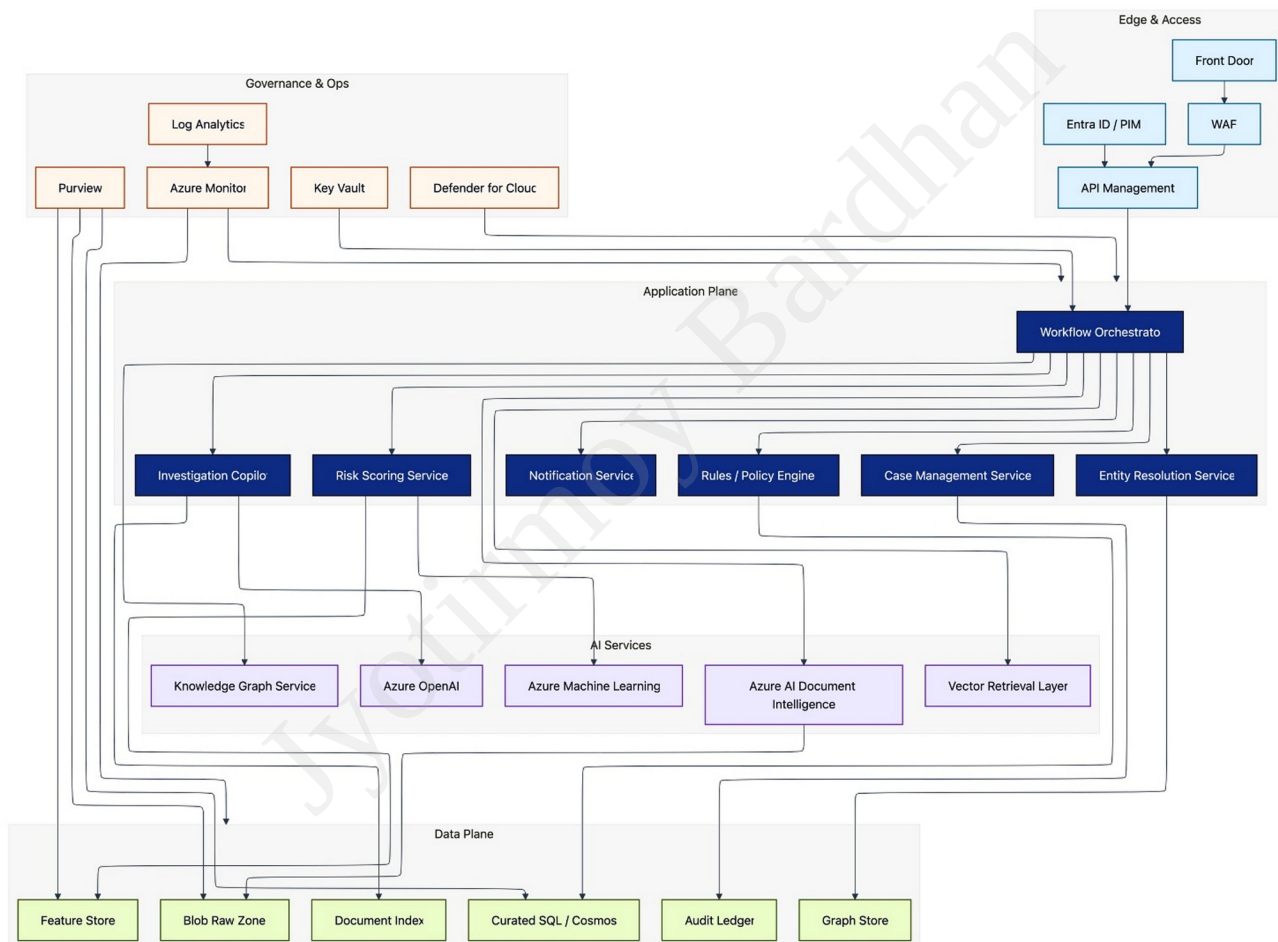
Model-Lifecycle



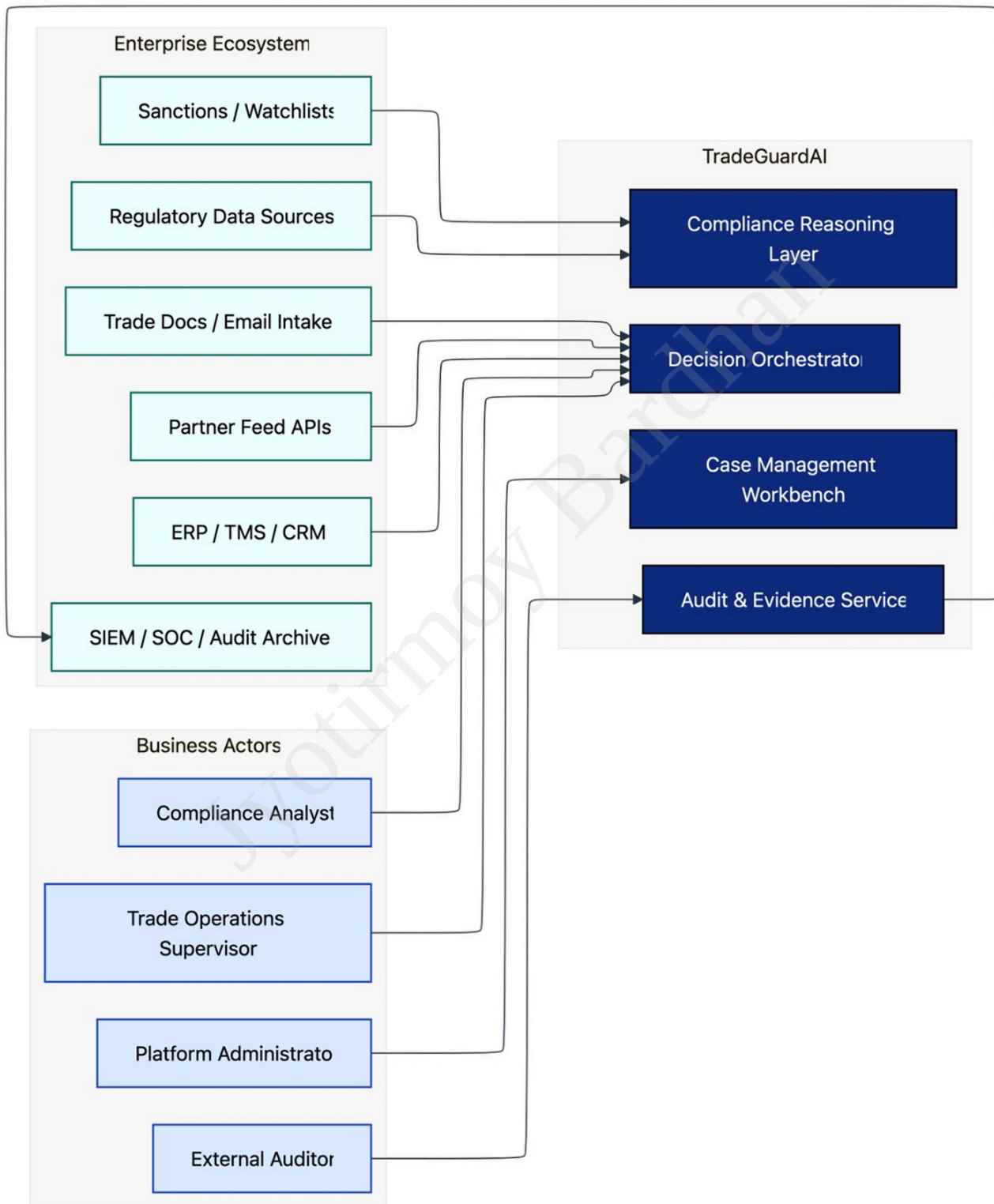
Data-Lineage



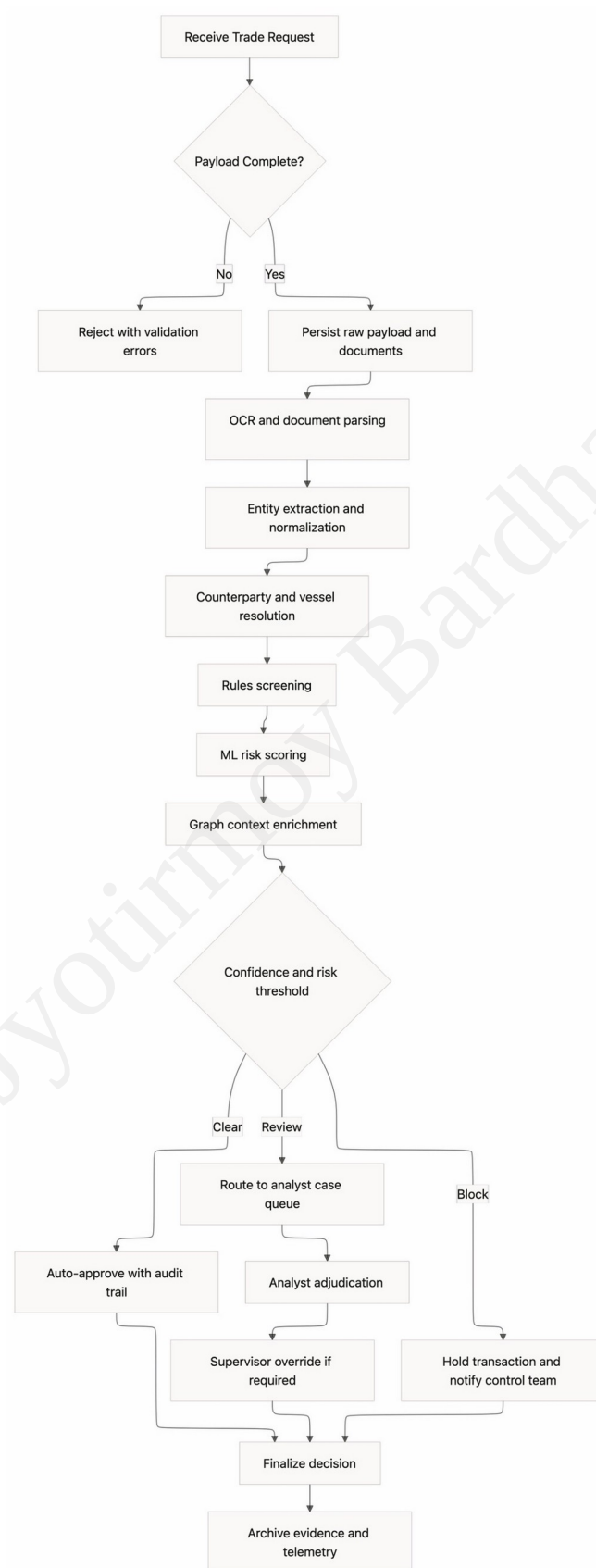
C4-Container



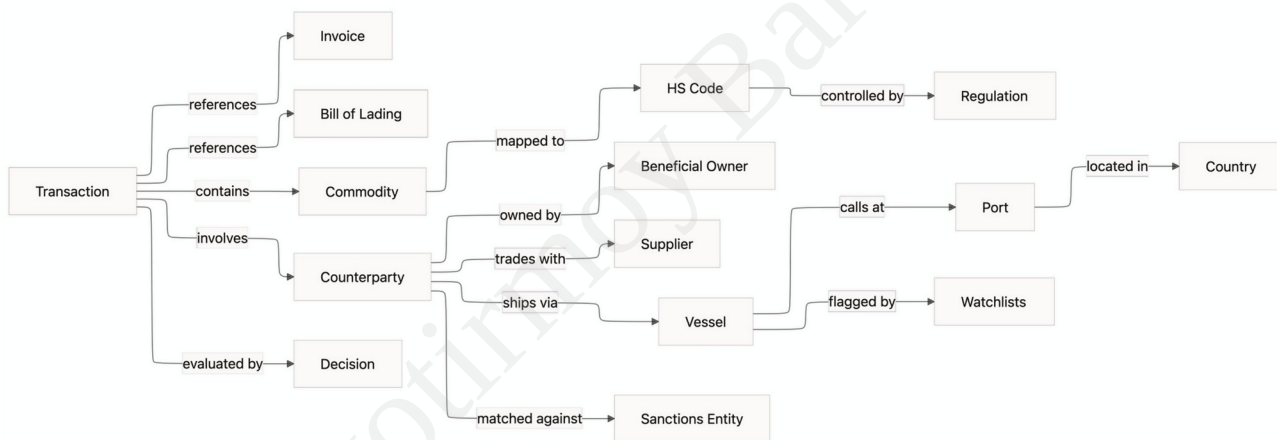
C4-Context



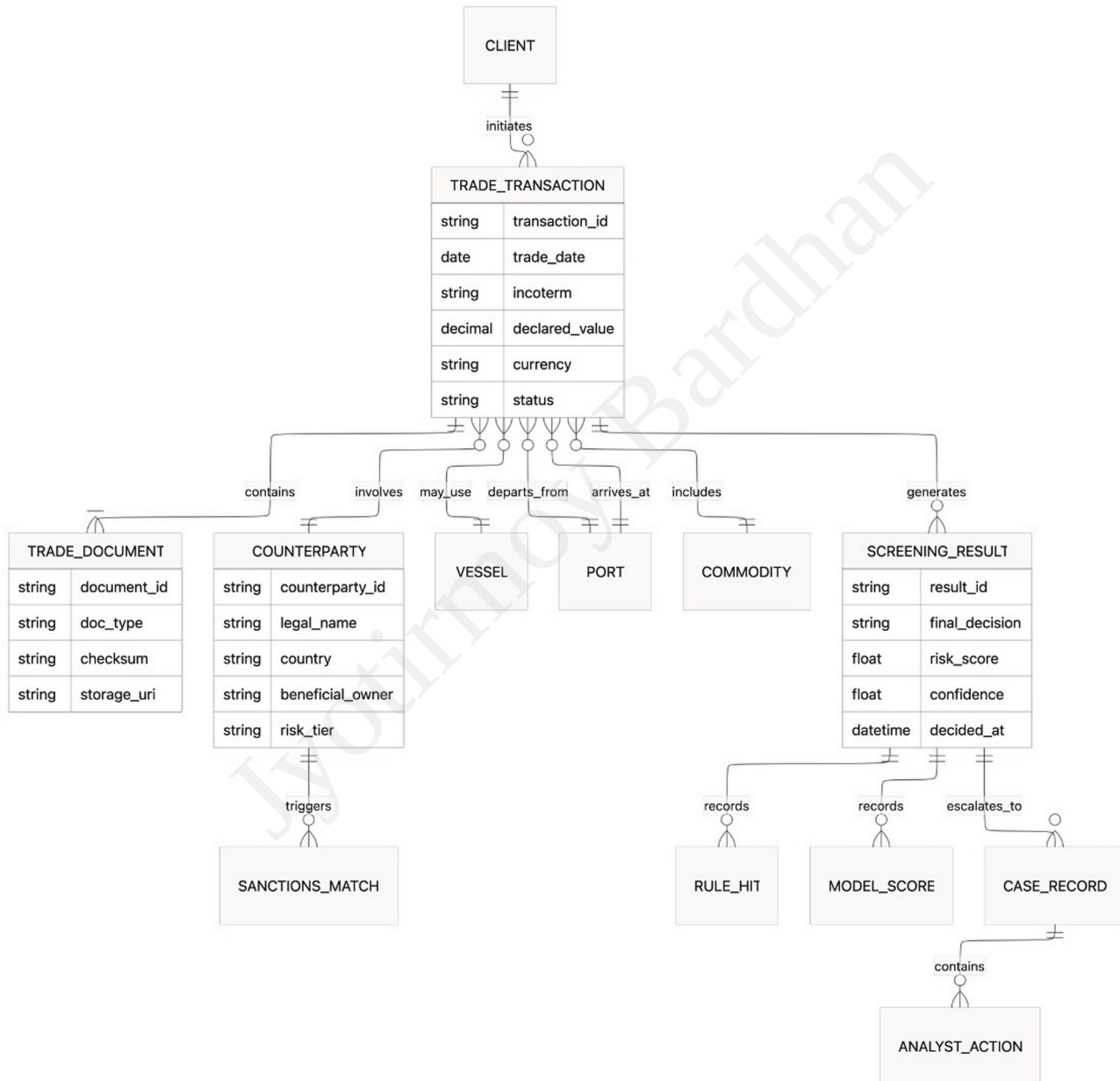
Workflow-Flowchart



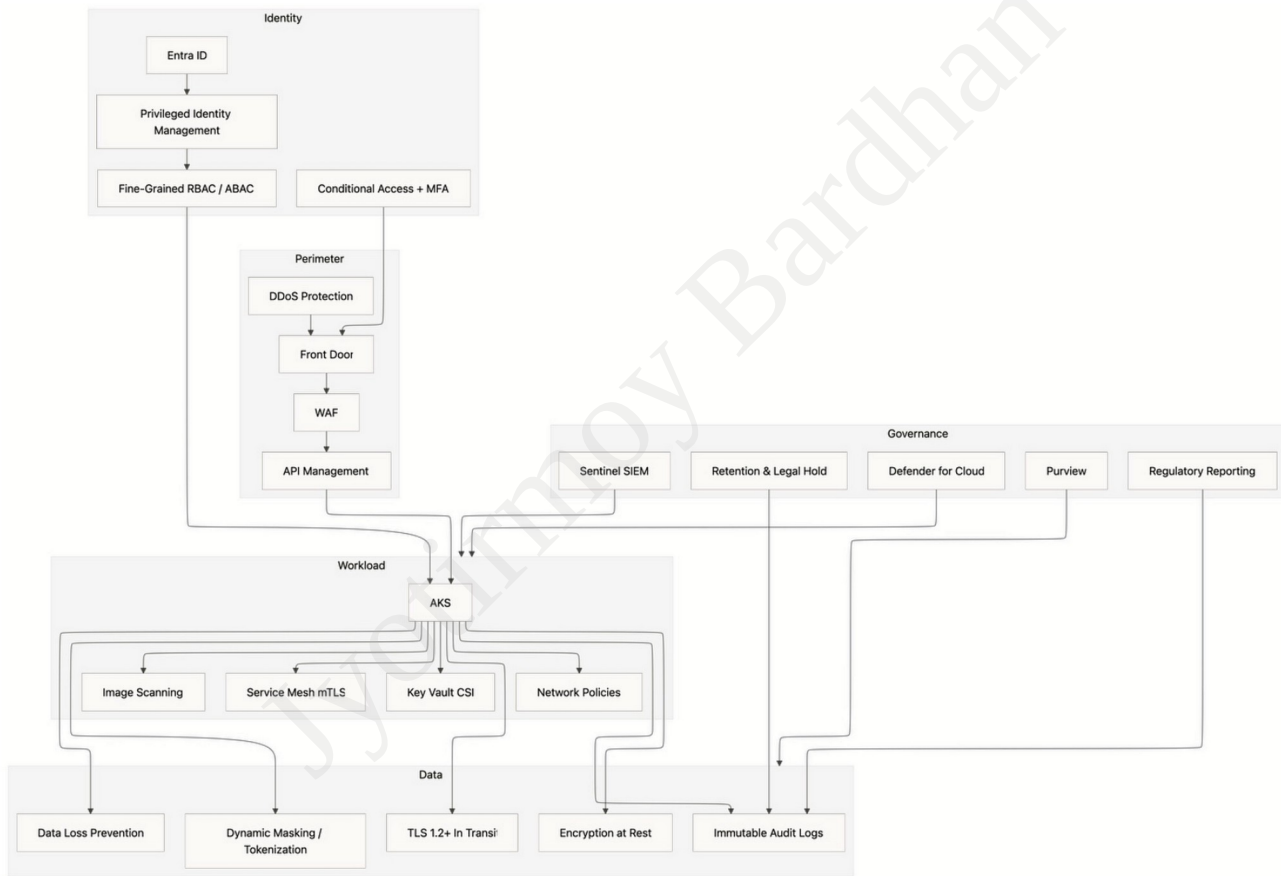
Knowledge-Graph



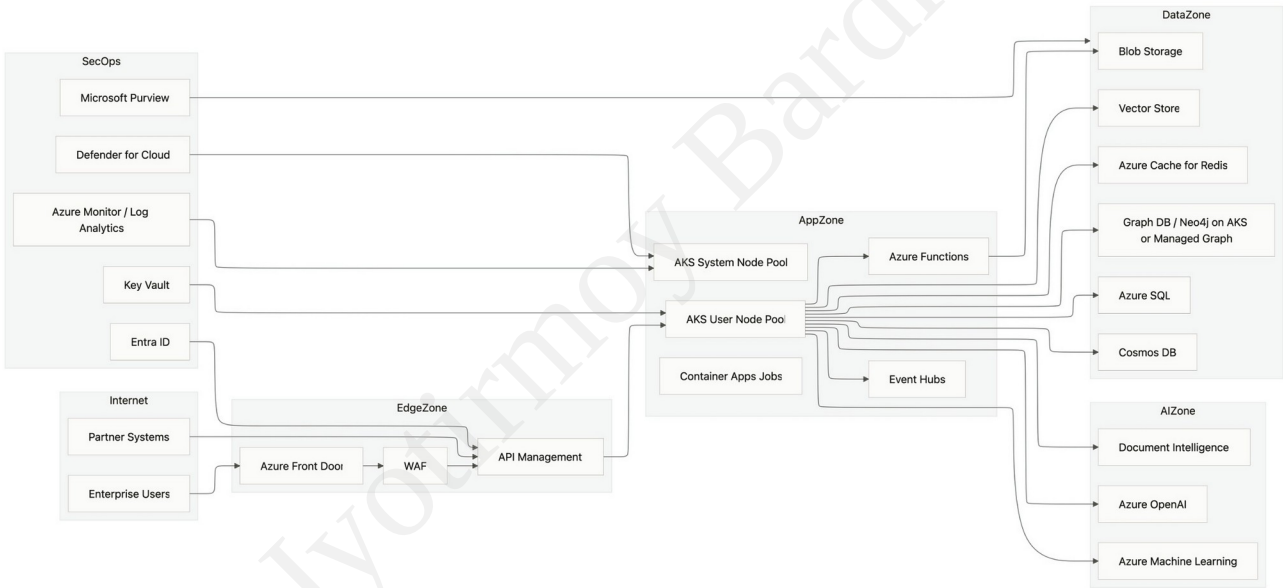
E-R-Model



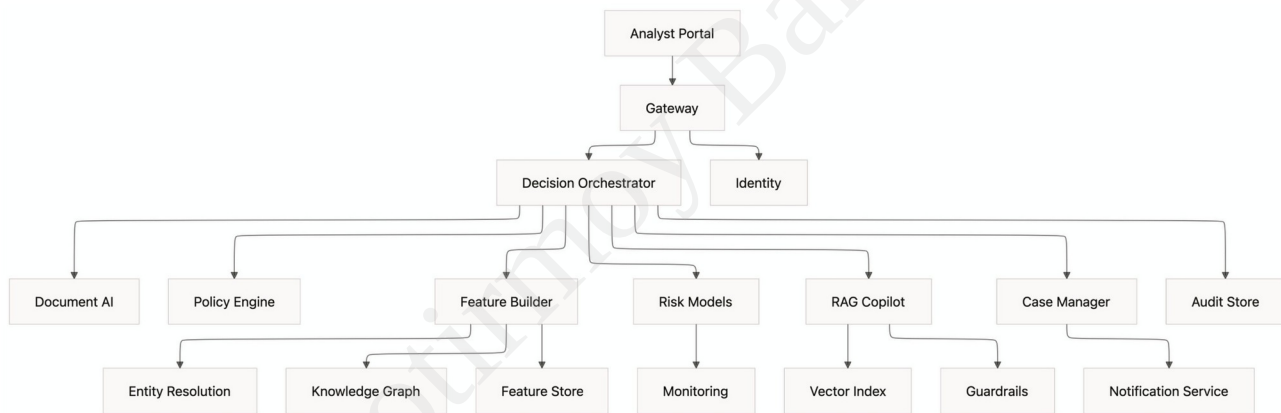
Security-Compliance-Architecture



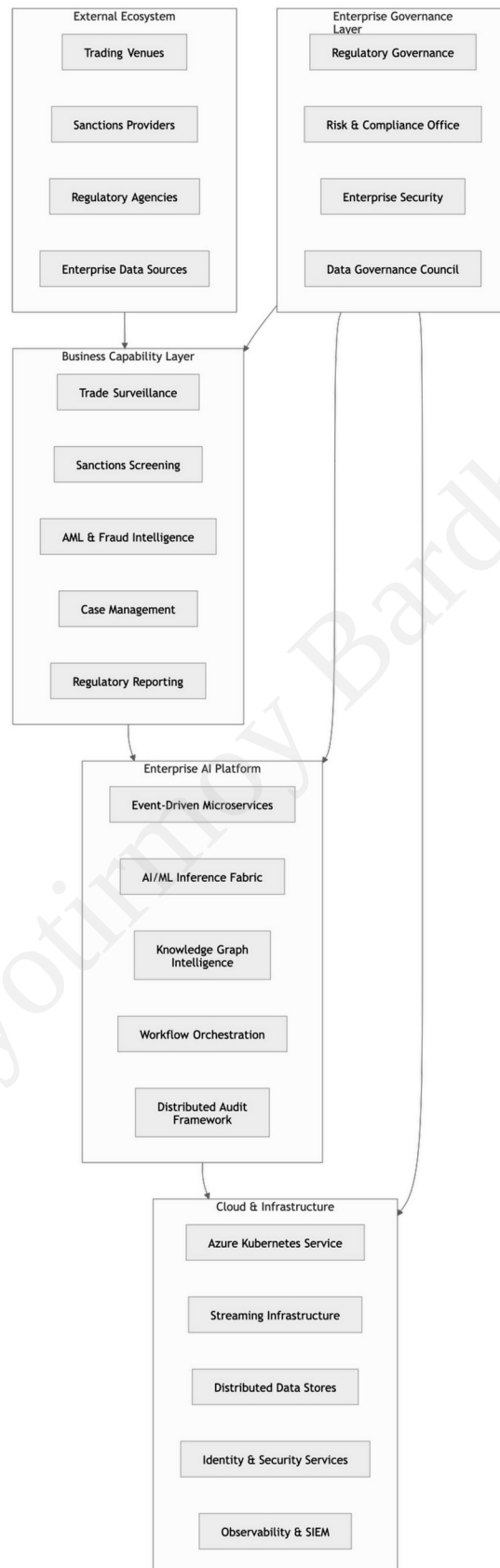
azure-deployment



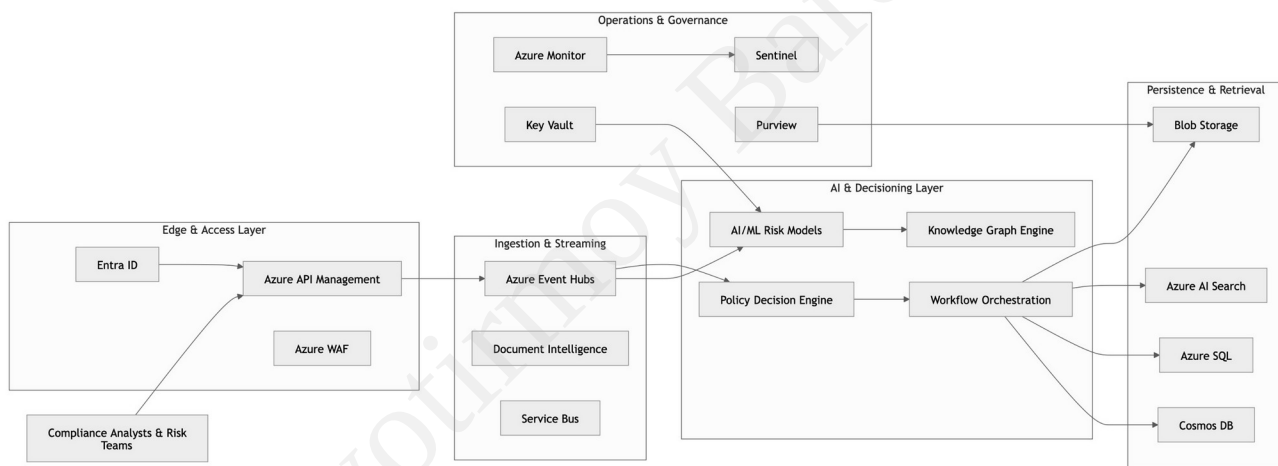
Dependency-Graph



Enterprise Architecture Diagram



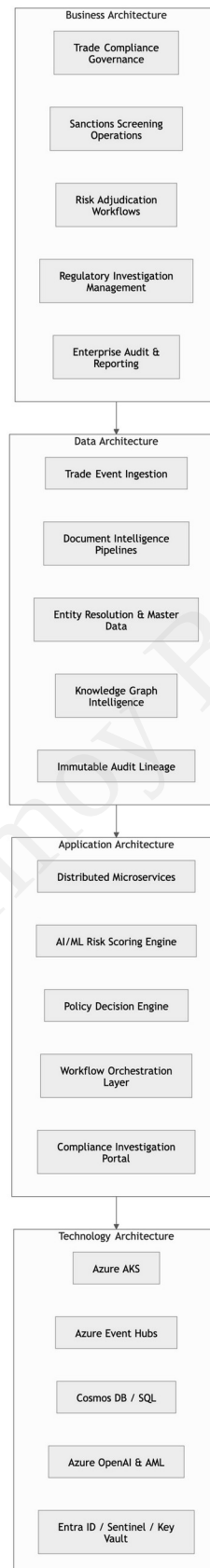
Solution Architecture Diagram



Zachman Framework Diagram



TOGAF Enterprise Architecture Diagram



TOGAF & Zachman Framework Implementation Mapping

TOGAF Implementation Across the Platform Architecture

The platform architecture was structured using TOGAF ADM-aligned enterprise architecture governance to ensure traceable alignment between regulatory objectives, business capabilities, distributed AI services, data governance, and cloud-native infrastructure controls. The Preliminary and Architecture Vision phases established the enterprise-wide compliance modernization strategy focused on sanctions intelligence, AML automation, trade surveillance, explainable AI decisioning, and immutable auditability. Business Architecture was implemented through domain capability decomposition covering: Trade Compliance Governance Sanctions Screening Operations Risk Adjudication Workflows Regulatory Investigation Management Enterprise Audit & Reporting Data Architecture implemented canonical trade event models, document intelligence pipelines, entity resolution services, graph-based contextual intelligence, lineage tracking, and evidence-retention controls. Application Architecture introduced event-driven distributed microservices, AI/ML inference orchestration, policy decision engines, workflow orchestration layers, compliance workbenches, RAG investigation copilots, and audit evidence services. Technology Architecture standardized Azure AKS, Event Hubs, Cosmos DB, Azure SQL, Azure OpenAI, Azure AI Document Intelligence, Key Vault, Sentinel SIEM, Front Door WAF, Purview governance, and multi-region disaster recovery topologies. TOGAF governance controls were embedded through architecture review boards, security baselines, CI/CD validation gates, drift monitoring, compliance traceability, and operational observability instrumentation.

Zachman Framework Realization Within the Solution Landscape

The Zachman Framework was operationalized to provide multi-dimensional traceability across enterprise stakeholders, architectural abstractions, operational workflows, and implementation artifacts. The contextual perspective (Planner Scope) defined: Business objectives Regulatory obligations Enterprise constraints Cross-border trade compliance requirements The conceptual perspective (Owner Model) mapped operational workflows including: Trade screening processes Risk escalation flows Human adjudication pathways Evidence retention obligations The logical perspective (Designer Model) formalized: Distributed system topology Canonical data models Security architecture Knowledge graph relationships Workflow orchestration semantics The physical perspective (Builder Model) translated the architecture into: Microservices decomposition Streaming infrastructure Containerized AI services Event-driven integration patterns AKS workload isolation boundaries The detailed implementation perspective operationalized: AKS deployment topologies CI/CD pipelines Infrastructure-as-Code controls Monitoring and telemetry frameworks Security hardening baselines The operational perspective incorporated: Incident response workflows Production observability Compliance reporting Operational governance controls Continuous assurance monitoring This Zachman-driven decomposition ensured architectural traceability between executive governance objectives, domain capabilities, distributed AI systems, operational workflows, and infrastructure implementation artifacts.